

MINI PROJECT

Small Office Network

Submitted by

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Submitted To

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BONAFIDE CERTIFICATE

Certified that this project report **“Small Office Network”** is the bonafide work of **“Ajay Kumar ”** who carried out the project work under my/our supervision.

Signature of the HoD

SIGNATURE

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With best Regards,

Ajay Kumar(23MCA20427)

PREFACE

This mini project, titled "*Small Office Network*," has been undertaken as a practical implementation of computer networking concepts using Cisco Packet Tracer. The primary objective of this project is to design and simulate a functional small office network that demonstrates efficient communication between multiple devices using core networking technologies.

In this project, a ring topology has been implemented using four Cisco 2811 routers, each connected to a switch. Every switch, in turn, connects to two PCs, ensuring that each workstation is part of the network. The design also includes essential office devices such as a server—configured as both a DNS and HTTP server—and a network printer, both integrated into the system via switches.

Routing Information Protocol (RIP) has been used to manage routing between the routers, ensuring dynamic and efficient path selection. All PCs in the network can communicate seamlessly with one another and can access services hosted on the DNS and HTTP server. The HTTP server hosts a sample index.html file, accessible through DNS resolution, demonstrating end-to-end service configuration and delivery.

This project not only reinforced theoretical knowledge but also provided hands-on experience with network planning, device configuration, and service deployment. It reflects the importance of proper design and routing in creating reliable and scalable network solutions for small office environments.

ABSTRACT

This project presents the design and implementation of a simulated *Small Office Network* using Cisco Packet Tracer. The network is structured in a ring topology comprising four Cisco 2811 routers, each connected to a switch, and each switch linked to two end-user PCs. Additional devices such as a printer and a server—configured to provide DNS and HTTP services—are integrated to replicate a functional office environment.

The Routing Information Protocol (RIP) is employed for dynamic routing between routers, enabling efficient communication and path discovery across the network. All PCs within the system are configured to communicate with each other and have access to network services, including domain name resolution and web browsing via the local DNS and HTTP server. The HTTP server hosts a simple index.html page to simulate a corporate homepage.

This project demonstrates a practical understanding of key networking concepts such as IP addressing, dynamic routing, server configuration, and inter-device communication. It provides a foundational example of how to build and manage a reliable network suited for small office environments.

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1. Introduction

In today's digitally connected world, reliable and efficient networking is a critical component of any organizational infrastructure. Small office networks, in particular, require well-structured communication systems to support day-to-day operations, including internet access, file sharing, printing, and internal communications. This project aims to simulate a small office network using Cisco Packet Tracer to provide a clear understanding of how various networking components interact within a professional environment.

The network is designed using a ring topology consisting of four Cisco 2811 routers. Each router is connected to a switch, and every switch is further connected to two PCs. Additionally, one switch is connected to a server that functions both as a DNS server and a web server (hosting an index.html file), while another switch is connected to a network printer. The use of the Routing Information Protocol (RIP) enables dynamic routing between the routers, allowing for effective data transmission and network scalability.

The goal of this project is to demonstrate how a small office setup can be designed to ensure seamless communication, centralized services, and resource sharing. By integrating DNS and HTTP services within the network, the simulation reflects a real-world office environment where users can resolve domain names and access internal web pages efficiently.

Through this project, key concepts such as routing protocols, IP addressing, device configuration, and server setup are explored and applied, providing hands-on experience in the field of computer networking.

2. Objective

The primary objective of this project is to design, implement, and simulate a functional small office network using Cisco Packet Tracer that demonstrates the integration and operation of essential networking components and services.

The specific objectives of the project are as follows:

1. **To design a ring topology network** using four Cisco 2811 routers, each connected to a switch, ensuring redundancy and continuous communication.
2. **To connect end-user devices** by linking two PCs to each switch, simulating a typical office environment with multiple workstations.
3. **To implement dynamic routing** using the Routing Information Protocol (RIP) for automatic route sharing and efficient data transmission between routers.
4. **To configure a DNS and HTTP server** that provides domain name resolution and hosts a basic web page (index.html), demonstrating internal web services.

5. **To integrate peripheral devices** such as a printer into the network, enabling resource sharing across connected PCs.
6. **To ensure full network communication**, allowing all PCs to interact with each other, access the server through DNS, and view the hosted web page via HTTP.
7. **To apply practical knowledge** of IP addressing, device configuration, and protocol management in a simulated environment, reinforcing core networking skills.

3. Tools & Technologies Used

The following tools and technologies were used in the development and simulation of the *Small Office Network* project:

- **Cisco Packet Tracer**
Used as the primary simulation tool for designing, configuring, and testing the network topology.
- **Cisco 2811 Routers**
Employed for routing between network segments using RIP (Routing Information Protocol).
- **Switches (Generic)**
Used to connect end devices such as PCs, servers, and printers within each local network.
- **End Devices**
 - **PCs** – For testing connectivity, DNS resolution, and web access.
 - **Server** – Configured with DNS and HTTP services.
 - **Printer** – Shared network resource accessible to PCs.
- **Routing Protocol**
 - **RIP (Routing Information Protocol)** – Used to enable dynamic routing among routers in a ring topology.
- **Networking Services**
 - **DNS Server** – Converts domain names to IP addresses.
 - **HTTP Server** – Hosts a sample web page (index.html) accessible to all network devices.
- **Protocols & Concepts**
 - IP Addressing (Static)
 - DNS Resolution
 - Dynamic Routing with RIP
 - Client-Server Architecture
 - Ring Topology Design

4. Code Implementation (Overview)

This section provides an overview of the key configuration commands used to implement the *Small Office Network* in Cisco Packet Tracer. The setup involves configuring routers, switches, PCs, a printer, and a server. Dynamic routing is enabled using RIP version 2 to ensure all devices across departments can communicate and access DNS and HTTP services hosted on the server.

Router Configurations

Each router is configured with:

- A FastEthernet interface (to switch and local network)
- One or two Serial interfaces (for ring connectivity)
- RIP routing for all connected networks

☒ Router0 Configuration	☒ Router1 Configuration
<pre>enable configure terminal hostname Router0 interface FastEthernet0/0 ip address 192.168.1.1 255.255.255.0 no shutdown interface Serial0/0/0 ip address 10.0.0.1 255.255.255.0 clock rate 64000 no shutdown interface Serial0/0/1 ip address 12.0.0.1 255.255.255.0 clock rate 64000 no shutdown router rip version 2 no auto-summary network 192.168.1.0 network 10.0.0.0 network 12.0.0.0</pre>	<pre>enable configure terminal hostname Router1 interface FastEthernet0/0 ip address 192.168.2.1 255.255.255.0 no shutdown interface Serial0/0/0 ip address 10.0.0.2 255.255.255.0 no shutdown interface Serial0/0/1 ip address 13.0.0.1 255.255.255.0 clock rate 64000 no shutdown router rip version 2 no auto-summary network 192.168.2.0 network 10.0.0.0 network 13.0.0.0</pre>

☒ Router2 Configuration <pre> enable configure terminal hostname Router2 interface FastEthernet0/0 ip address 192.168.3.1 255.255.255.0 no shutdown interface Serial0/0/0 ip address 12.0.0.2 255.255.255.0 no shutdown interface Serial0/0/1 ip address 11.0.0.2 255.255.255.0 no shutdown router rip version 2 no auto-summary network 192.168.3.0 network 12.0.0.0 network 11.0.0.0 </pre>	☒ Router3 (Assumed 'Router4' in your diagram) <pre> enable configure terminal hostname Router3 interface FastEthernet0/0 ip address 192.168.4.1 255.255.255.0 no shutdown interface Serial0/0/0 ip address 13.0.0.2 255.255.255.0 no shutdown interface Serial0/0/1 ip address 11.0.0.1 255.255.255.0 clock rate 64000 no shutdown router rip version 2 no auto-summary network 192.168.4.0 network 13.0.0.0 network 11.0.0.0 </pre>
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PC Configuration (for each PC)

Department 1 (Router0, Switch0) • PC1 (IP: 192.168.1.2) ◦ IP Address: 192.168.1.2 ◦ Subnet Mask: 255.255.255.0 ◦ Default Gateway: 192.168.1.1 (Router0) ◦ DNS Server: 192.168.4.100 (Server) • PC2 (IP: 192.168.1.3) ◦ IP Address: 192.168.1.3 ◦ Subnet Mask: 255.255.255.0 ◦ Default Gateway: 192.168.1.1 (Router0) ◦ DNS Server: 192.168.4.100 (Server)	Department 2 (Router1, Switch1) • PC3 (IP: 192.168.2.2) ◦ IP Address: 192.168.2.2 ◦ Subnet Mask: 255.255.255.0 ◦ Default Gateway: 192.168.2.1 (Router1) ◦ DNS Server: 192.168.4.100 (Server) • PC4 (IP: 192.168.2.3) ◦ IP Address: 192.168.2.3 ◦ Subnet Mask: 255.255.255.0 ◦ Default Gateway: 192.168.2.1 (Router1) ◦ DNS Server: 192.168.4.100 (Server)
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Department 3 (Router2, Switch2) • PC5 (IP: 192.168.3.2) ◦ IP Address: 192.168.3.2 ◦ Subnet Mask: 255.255.255.0 ◦ Default Gateway: 192.168.3.1 (Router2) ◦ DNS Server: 192.168.4.100 (Server) • PC6 (IP: 192.168.3.3) ◦ IP Address: 192.168.3.3 ◦ Subnet Mask: 255.255.255.0 ◦ Default Gateway: 192.168.3.1 (Router2) ◦ DNS Server: 192.168.4.100 (Server)	Department 4 (Router3, Switch3) • PC7 (IP: 192.168.4.2) ◦ IP Address: 192.168.4.2 ◦ Subnet Mask: 255.255.255.0 ◦ Default Gateway: 192.168.4.1 (Router3) ◦ DNS Server: 192.168.4.100 (Server) • PC8 (IP: 192.168.4.3) ◦ IP Address: 192.168.4.3 ◦ Subnet Mask: 255.255.255.0 ◦ Default Gateway: 192.168.4.1 (Router3) ◦ DNS Server: 192.168.4.100 (Server)
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Printer Configuration

- **IP Address:** 192.168.2.100
- **Subnet Mask:** 255.255.255.0
- **Gateway:** 192.168.2.1

Server Configuration

- **IP Address:** 192.168.4.100
- **Subnet Mask:** 255.255.255.0
- **Gateway:** 192.168.4.1

Services Enabled:

- **DNS:**
 - Domain: `www.Ajayprofile.com`
 - IP: 192.168.4.100
- **HTTP:**
 - Hosts `index.html`, accessible by all PCs via the DNS name.

5. Result and Analysis

Result

The *Small Office Network* was successfully configured. The following outcomes were achieved:

1. Ping Tests: All PCs communicated with each other across departments.

Laptop1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:

Reply from 192.168.4.2: bytes=32 time=28ms TTL=125
Reply from 192.168.4.2: bytes=32 time=17ms TTL=125
Reply from 192.168.4.2: bytes=32 time=24ms TTL=125
Reply from 192.168.4.2: bytes=32 time=28ms TTL=125

Ping statistics for 192.168.4.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 28ms, Average = 24ms
```

Laptop5

Physical Config Desktop Programming Attributes

Command Prompt

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 21ms, Maximum = 30ms, Average = 26ms

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=30ms TTL=125
Reply from 192.168.2.3: bytes=32 time=23ms TTL=125
Reply from 192.168.2.3: bytes=32 time=32ms TTL=125
Reply from 192.168.2.3: bytes=32 time=30ms TTL=125

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 32ms, Average = 28ms
```

2. DNS Resolution: PCs resolved the domain www.Ajayprofile.com to the server's IP (192.168.4.100).

Laptop0

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 192.168.4.100

Server0

Physical Config Services Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DNS

DNS Service: ☒ On ☐ Off

Resource Records

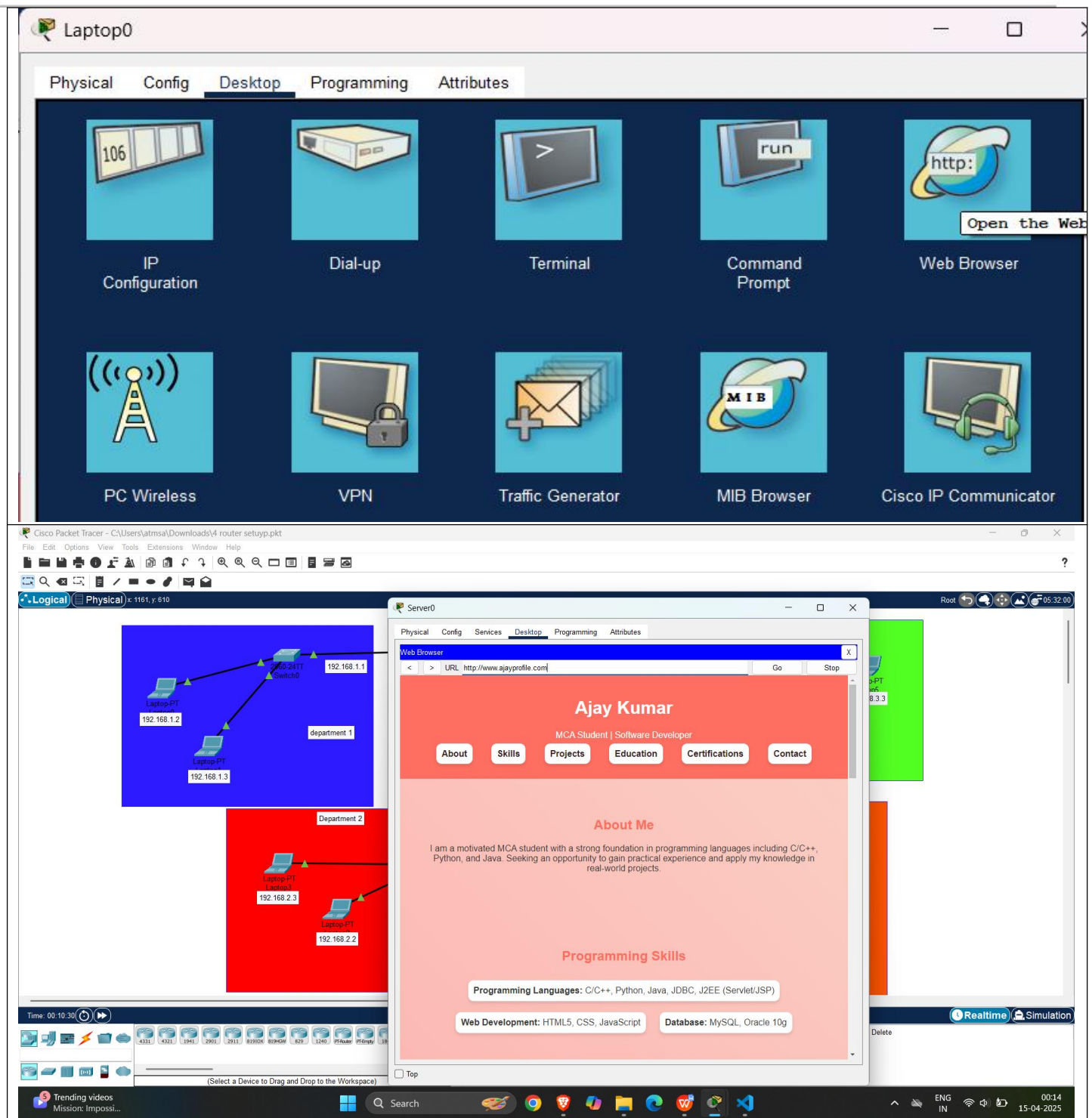
Name: Type: A Record

Address:

Add Save Remove

No.	Name	Type	Detail
0	www.ajayprofile.com	A Record	192.168.4.100

3. HTTP Access: PCs accessed the index page hosted on the server.



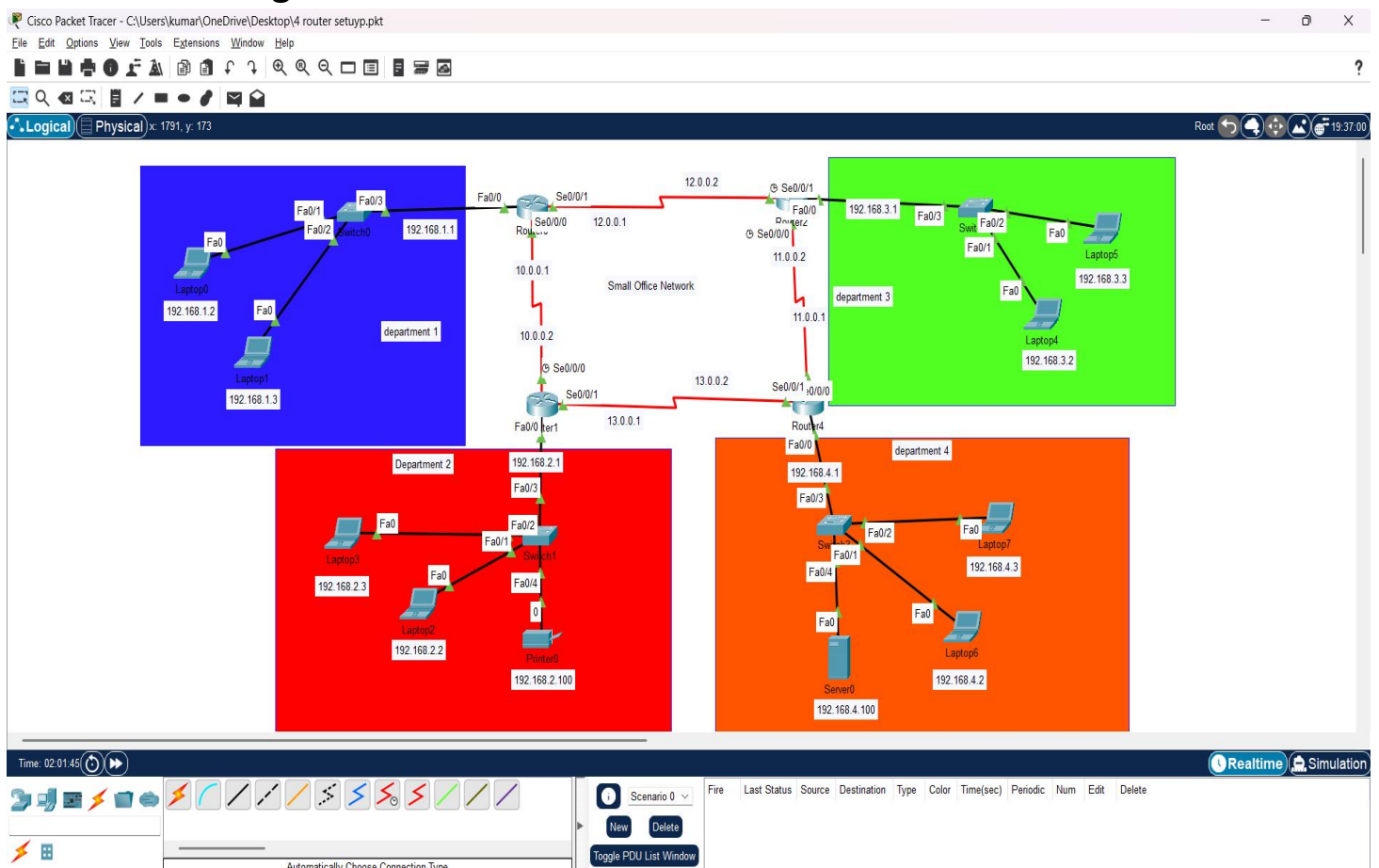
The screenshot displays the Cisco Packet Tracer interface. At the top, a 'Laptop0' window shows a 'Desktop' tab with icons for IP Configuration, Dial-up, Terminal, Command Prompt, Web Browser, PC Wireless, VPN, Traffic Generator, MIB Browser, and Cisco IP Communicator. Below this, the main workspace shows a network topology with two departments: Department 1 (blue background) and Department 2 (red background). Department 1 contains a switch (192.168.1.1) connected to two laptops (192.168.1.2 and 192.168.1.3). Department 2 contains two laptops (192.168.2.2 and 192.168.2.3). A 'Server0' window is open, showing a web browser at 'http://www.ajayprofile.com'. The website displays the profile of 'Ajay Kumar', an MCA Student and Software Developer, with sections for About Me, Programming Skills, and Contact. The bottom status bar shows the time as 00:10:30 and the date as 15-04-2025.

4. **Printer Access:** The printer in Department 2 was accessible to PCs in that department.

Analysis

- **RIP Protocol:** Successfully propagated routing information, ensuring inter-department communication.
- **Subnetting:** Correctly segmented the network, allowing smooth communication within and between departments.
- **Server Services:** DNS and HTTP services functioned as expected, with all PCs able to resolve names and access the web page.
- **Ring Topology:** The ring topology provided redundancy, ensuring the network remained functional even if one link failed.

6. Network Design



7. Conclusion

The *Small Office Network* project successfully demonstrated the design and implementation of a functional, multi-departmental network using Cisco Packet Tracer. By configuring routers in a ring topology with RIP routing, and integrating switches, PCs, a printer, and a server with DNS and HTTP services, the network achieved full connectivity and service access across all departments. This setup illustrates key networking concepts such as subnetting, dynamic routing, DNS resolution, and inter-device communication in a real-world office scenario. The project meets its objectives and provides a scalable and efficient network model for small office environments.

8. References

1. Cisco Networking Academy – *Introduction to Networks*
2. Packet Tracer Official Documentation – Cisco Systems
3. RFC 1058 – *Routing Information Protocol*
4. Wikipedia – *RIP (Routing Information Protocol)*
5. Youtube Channel – <https://www.youtube.com/watch?v=i83qrFq3BYQ>